

Art as an Algorithmic Virus: Unifying the Generative Crash and AI Value Convergence via Cognitive Affordances

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Abstract—Generative AI inherently triggers a computational failure mode in human observers (a “generative crash”) due to a lack of latent intentionality required for Inverse Reinforcement Learning (IRL) convergence. Artistic appreciation operates as the biological execution of this IRL process. To address the generative crash and broader AI alignment failures, I introduce the Ghost Scale (an HCI cognitive affordance for identifying intentionality) and propose Cooperative Inverse Reinforcement Learning (CIRL) to mimic biological value transmission. The *Intent Extraction Limit* is formalized to define the prior relationship. Applying this model addresses two major issues: generative AI’s friction with the art community (via the Ghost Scale, a cognitive affordance and UX framework for signaling intentionality) and AI alignment [via a proposed shift from Reinforcement Learning from Human Feedback (RLHF) toward top-down value capture through Cooperative Inverse Reinforcement Learning (CIRL) informed by world models]. Six empirical hypotheses are proposed to test the framework.

Index Terms—AI Alignment, Active Inference, Inverse Reinforcement Learning, Generative AI, Cognitive Affordances, Human-Computer Interaction, Free Energy Principle

I. INTRODUCTION

To align artificial intelligence with human aesthetics, we must first strip art of its mysticism and reduce it to measurable structural telemetry: art is *compressed intent*. Under this biological model, emotional resonance and structural emergence are not accidents; they are the successful somatic decoding of a creator’s nested, subconscious priors [1]. Therefore, *art* is simply any artifact into which a cognitive agent concentrates a massive density of thermodynamic resources and decision-making to serve a goal. Artifacts with more decisions are more artful. Domain is irrelevant: a painting, a bridge, a navigational route, and a smile all execute the exact same biological process and are judged as more or less artful depending on the concentration of decisions made in their creation.

This definition accommodates apparent counterexamples from art history. Duchamp’s *Fountain* (1917) was a mass-produced urinal submitted as art, and involved minimal physical fabrication but concentrated enormous decision-density into a single act of selection, contextualization, and institutional provocation [2]. Under the present framework, the artfulness resides not in the manufactured object but in the hierarchical compression of conceptual decisions that Duchamp’s act encoded. Similarly, aleatory composition and process-based art deliberately remove low-level decisions, but the meta-decision

to constrain or relinquish control itself carries high intentional density. The framework thus distinguishes between the absence of decisions (low artfulness) and the deliberate, structured removal of decisions (potentially high artfulness), a distinction that maps naturally onto the hierarchical bundling described by automaticity.

I will define *appreciation* as: identifying the actions used to create an artifact, using that information to derive the goals of the creator via Inverse Reinforcement Learning (IRL) [3], and using the alignment of those inferred goals with the viewer’s own values to determine the magnitude of the resulting synaptic update. Within this biological framework, the observer’s intent-extraction protocol is modeled at the computational level as Maximum Entropy Inverse Reinforcement Learning (MaxEnt IRL) [4], assuming the creator maximized reward while maintaining maximum behavioral entropy. In other words, appreciation is the act of inferring a maker’s intent, learning to mimic the (expectedly) efficient steps that maker took to achieve that intent, and then halting the process if the inferred intent diverges from the viewer’s.

We are not passive recipients of artistic inputs. We actively engage with the world through the lens of intentionality [5].

The Theory of Mind network we use for modeling the perspectives of others is notably active when we engage with something we are told is “art” [6], [7]. Behavioral evidence directly supports this prediction. Messingschlager and Appel (2025) demonstrated that labeling an artifact as AI-generated reduces appreciation, and crucially that this effect is mediated by the perceived agency and experience ascribed to the creator (both Theory of Mind constructs) [8]. Their finding that mind perception statistically mediates the appreciation gap is consistent with the present framework’s prediction that the devaluation of generative art is driven by a failure to converge upon a coherent latent goal rather than by surface-level aesthetic judgment; the neural mechanism of this deactivation is tested directly in H1.

To address this, I will propose the **Ghost Scale**, a UX framework that acts as a cognitive affordance for the viewer in the identification of intentionality. This scale, properly deployed, would also be effective in both the immediate avoidance of model collapse [9] and eventual AI alignment (see H5).

Cultural transmission of values is already optimized for the

biological execution of inverse reinforcement learning through the intent extraction of top-down precision weighting [10]. Currently, we train AI by extracting value information from bottom-up aesthetic capture mechanisms using reinforcement learning from human feedback (RLHF) [11]. This will inevitably result in the creation of shallow output in terms of both objective artistic assessment and the hierarchical alignment with human values. We should instead take advantage of this widespread human behavior and capability by utilizing existing neurological and cultural architecture to capture appropriate value information through Cooperative Inverse Reinforcement Learning (CIRL) [12] and subsequently using it to inform convergence of world models such as GENIE and Joint Embedding Predictive Architectures [13].

Human brains optimize to extract intention from their environment already. Addressing alignment requires not just coding better guardrails, but learning to capture humanity’s constant broadcasting of value and intent information.

II. THE NEUROBIOLOGY OF INTENT EXTRACTION

Analysis begins with physical artifacts. To process the language of intent, the viewer’s neocortex constructs a possible physical and cognitive trajectory for the creation of an object through the process of *embodied simulation* [14]. Mirror neurons are hypothesized to contribute bottom-up error correction for the top-down theory of mind network [15], which together map inferred behaviors and their physical deviations onto the viewer’s own visceromotor and somatosensory architecture.

Under the **Thousand Brains Theory** [16], this parsing is distributed across tens of thousands of independent cortical columns. Deep visual processing of an artifact necessitates that these columns functionally simulate the kinetic physics required for execution. This spatial mapping extends to abstract, higher-order concepts. Evidence from fMRI studies on the entorhinal cortex demonstrates that humans organize abstract conceptual knowledge using a grid-like code, effectively mapping non-spatial cognitive trajectories onto geometric coordinate systems [17]. Therefore, when observing a programmer manipulate data architectures, the observer’s embodied simulation still fires; the Theory of Mind network maps these abstract problem-solving strategies onto conceptual reference frames, error-correcting the inferred latent goal against the observer’s own spatial and cognitive geometry.

However, another source of error correction for the simulation of higher order concepts is the mirror neuron system simulating internal states (i.e., *affective empathy*) [18]. Partially through an equivalent error correction through conceptual grid space and partially through the use of lower-order data from the non-grid architecture as a precision weighting mechanism for cortical prediction errors, bottom-up processing gives error correction through empathy in the simulation of the creation of the artifact. Through both top-down processing via *cognitive empathy* in the Theory of Mind network and bottom-up processing via *affective empathy* in the limbic system and Mirror Neuron System, we work to minimize free energy [19]

and converge upon a cohesive model that explains the creation of that artifact.

A. The Simulation Firewall of Disgust

The **Default Mode Network** operates as a primary framework for simulation and is a core integration hub for the Theory of Mind [20]. While simulating, the amount of update of neural weights is determined by the plasticity gating mechanism governed by the **Salience Network** [21]. This network, when active, encourages the update of neural weights during simulation. Epistemic disgust moderates this process, triggered by a large value misalignment in the converged model.

The calculation of the value disagreement itself implies simulation, which inherently drives Hebbian learning due to gating imperfections [22]. This forced Hebbian learning is likely the mechanism for indoctrination and propaganda, and the gating mechanism of disgust that resists these effects will be reflected in the first term of the **intent extraction limit**.

III. THE INTENT EXTRACTION LIMIT

I formally define artistic appreciation (Ψ) as the magnitude of systemic prediction error reduction achieved via the synaptic internalization of a conspecific’s behavioral heuristics. This integration operates under the Free Energy Principle [19], where the observer executes what is formally equivalent to Inverse Reinforcement Learning (IRL) to reverse-engineer a hidden reward function. This equivalence is not stipulated as analogy but derived from modeling both agents under the FEP (see Appendix A.1). The process is governed by metabolic constraints and value alignment.

The magnitude of this synaptic update is driven by Bayesian surprise, quantified as the Kullback-Leibler divergence [23] between the observer’s prior expectation of human motivation [$P(R)$] and the updated posterior extracted from the physical artifact [$P(R \mid \tau)$]. Because biological neural networks chunk information to manage working memory constraints, this integration is bounded by the algorithmic cognitive load. Using Variational Inference (VI), this friction is expressed as the minimization of the Kullback-Leibler divergence between the observer’s biological prior (a flat entropy distribution representing baseline metabolic constraints) and the updated posterior extracted from the artifact.

Integration is catalyzed by epistemic trust (κ), evaluated under Zahavian signaling constraints based on unforgeable production costs [24]. The logarithmic amplifier $-\ln(1 - \kappa)$ was selected for three converging reasons. First, it captures the empirically observed nonlinear scaling of interpersonal and automation trust, where incremental evidence near certainty produces disproportionate behavioral commitment [25]. Second, it enforces the biologically necessary constraint that κ can never reach 1: trust is asymptotic, never absolute, and the logarithmic barrier prevents the integration multiplier from diverging to infinity. Third, $-\ln(1 - \kappa)$ is the information-theoretic self-information of untrustworthiness, scaling the gate in units commensurate with the KL divergence term it modulates. If this asymptotic bound is artificially bypassed,

the brain is forced to hallucinate intent to resolve the resulting computational paralysis (see Appendix C, H6).

The brain’s active inference mechanisms operate under a hard metabolic constraint [19], [26], modeled here as an autonomic sigmoidal gating function that aborts inference when expected information gain falls below a metabolic threshold. The sigmoidal form is standard in computational neuroscience for modeling neural population threshold responses, where a continuous input is converted to a binary activation decision via accumulated evidence [27]. The gain parameter k reflects the meta-precision of the gating function itself (the precision with which the brain weighs sensory precision against the disgust threshold) [27]. This gate balances bottom-up *sensory precision weighting* (ω) against a dynamic threshold of epistemic disgust ($\theta_{E,C}$). This threshold scales inversely with the organism’s available metabolic reserves (E) and directly tracks the alignment of the artifact with the observer’s survival priors (C), functioning as the foundational reward against which the creator’s inferred state (R_c) is measured:

$$\theta_{E,C}(\kappa) = \theta_{base}(E) + \lambda D_{KL}(Q(R | \tau) \parallel P_c(R))$$

The complete formalization of the Intent Extraction Limit is therefore expressed as:

$$\Psi = \left[\frac{1}{1 + e^{-k(\omega - \theta_{E,C}(\kappa))}} \right] [-\ln(1 - \kappa)] \\ \times D_{KL}(Q(R | \tau) \parallel P_0(R))$$

Computationally, within latent diffusion architectures, the causal link between intent and physical execution is severed, corresponding to the collapse of idiosyncratic epsilon-prediction errors. Simultaneously, the epistemic disgust threshold $\theta_{E,C}(\kappa)$ acts as an autonomic regularization penalty preventing weight updates from misaligned latent structures.

This framework constrains the mechanics of memetic transmission to two distinct biological failure modes:

A. The *Epistemic Disgust Firewall* ($\Psi \rightarrow 0$):

If the creator’s inferred goal radically diverges from the observer’s fundamental values ($P_c(R)$), the ideological divergence term $D_{KL}(Q(R | \tau) \parallel P_c(R))$ spikes. This drives the disgust threshold $\theta_{E,C}$ to mathematically override the trust variable κ . The sigmoid evaluates to near zero, closing the gate. The brain may recognize the aesthetic complexity of the artifact, but it autonomously blocks Hebbian integration to protect its own network weights from a misaligned memetic signature.

B. The *Generative Crash* ($L(R) \rightarrow \infty$):

When processing synthetic media generated by latent diffusion architectures [28], idiosyncratic deviations mathematically collapse because structural execution is decoupled from a biological motor-policy. While the artifact possesses high Shannon entropy (visual complexity) [29], it lacks the high Kolmogorov complexity (description length) inherent to biological constraint satisfaction [30], [31]. Because the variance

is stochastic rather than intentional, the biological prior $P_0(R)$ cannot map the output to a coherent human hidden state, and the algorithmic cognitive load required to model this unconstrained space diverges toward infinity ($L(R) \rightarrow \infty$). The biological MaxEnt IRL calculation terminates, defaulting the extracted reward to zero ($R_{inferred} = 0$) to conserve finite computational resources.

IV. AI ALIGNMENT IMPLICATIONS

The preceding biological mechanisms suggest (but do not formally entail) specific implications for computational alignment. The following structural correspondences are proposed as extensions of the framework, contingent on assumptions about the scalability of CIRL architectures and the capacity of world models to capture hierarchical intent. The biological MaxEnt IRL executed during appreciation is derived from the Free Energy Principle directly (see Appendix A.1) and maps to the inverse of next-token prediction in language models. Embodied simulation maps functionally to the forward passes of world models. The second correspondence is treated as a computational-level analogy; the first is a derived consequence of the shared theoretical substrate.

AI alignment will require extracting value data, using it to model humanity’s higher-order values, and then turning that resulting value information into appropriately aligned action. I will address only the first two issues here, starting with the second.

A. *Modeling R*

The prevailing methodology of reinforcement learning is optimizing for low-level sensory data. RLHF optimizes a single scalar bound to cognitively fatigued raters prioritizing surface heuristics (e.g., formatting, tone, superficial accuracy) over alignment with higher-order latent values (e.g., helpfulness, honesty, compassion) [32].

Unfortunately, interacting with automated output turns off our theory of mind network and higher-order reasoning. By definition, we are not evaluating, rating, or providing any input on the axis of what humanity truly seems to value: the minimization of $D_{KL}(Q(R | \tau) \parallel P_0(R))$ (i.e., evidence of efficient, effective, and novel decisionmaking; or *artfulness*). Instead, we optimize for *aesthetics*: the combination of base heuristics of attractiveness and the initial impressions of depth (κ) before the Intent Extraction Limit converges.

Next-token prediction and latent diffusion are explicitly mapping what can be observed: $P(\tau_{t+1} | \tau_{1:t})$. We are not even capable of identifying the latent cause (R), but the information is being put out into the world constantly. Architectures such as Google’s GENIE [33] and Meta’s Joint-Embedding Predictive Architectures [13] work to decode the higher-order reasoning that would be necessary to isolate and align with humanity’s R . However, world models of all kinds are important explorations into this space. I argue that these architectures are fundamentally required for identification and alignment with humanity’s latent (R).

B. Extracting Value Data

However, possessing the architecture to model does not solve the sycophancy loop of RLHF [34]. To align the machine, we must utilize **Cooperative Inverse Reinforcement Learning** [12]. Within a CIRL topology, the interaction is framed as a cooperative game where the human reward function remains explicitly hidden from the artificial agent. This mathematical constraint ensures the system operates under permanent epistemic uncertainty, forcing it to utilize observed behavioral trajectories and cultural artifacts (τ) as evidentiary telemetry to update its probability distributions of human intent, mimicking the biological transmission of values.

When using this method, we are recreating humanity's native language for the appreciation of art via inverse reinforcement learning and actively using it to extract value data that humanity is not just passively emitting but actively signaling.

This opens the door for a potential step towards a solution: one that reduces the environmental epistemic uncertainty κ while helping humans with the cognitive load of calculating a need for epistemic disgust and generative crashes.

To solve both problems, we need a cultural meta-signal for degrees of AI-generated content. We need a cognitive affordance for both human minds and AI training. Humans may conserve energy with such a signal by quickly triggering a generative crash and climbing into higher-order reasoning (such as guessing a prompt's motivation). This would allow the viewer to save energy calculating $\theta_{E,C}$ by quickly feeding the user the answer of a low-value κ . The epistemic uncertainty inherent in CIRL may also be reduced through the same method.

The design requirements for the affordance are that it be unobtrusive-yet-visible, intuitively learned, exist in a largely uncontested design space, be self-enforcing by design, robust against future technological development, and simple to apply. Most of all, it must be appealing to artists (defined as "having depth of reasoning behind the decision process").

V. THE GHOST SCALE

The Ghost Scale follows an intuitive framework: in any artifact with text, the main text color is the color of human intent. In that same artifact, the main background color is the color of output that is devoid of higher-order reasoning. For example, in Figure 1, black text denotes human origin while white text is exclusively from an AI with minimal selection (if any).

This may be expressed as percentages of image opacity. Lower levels of image opacity are a relatively sparse region in the latent design space because they actively degrade *perceptual fluency*. In addition to following WCAG contrast constraints [35], text at 60% opacity reduces the luminance contrast ratio to actively degrade perceptual fluency. Rather than lowering the thermodynamic cost of reading at the ocular level, this imposes a marginal perceptual tax.

However, within an active inference framework, this deliberate degradation of sensory precision acts as an autonomic

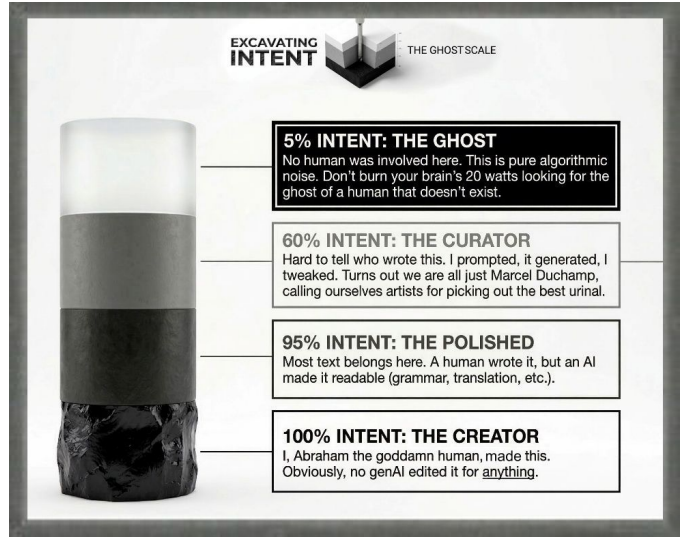


Fig. 1. AI-Generated Image, curated from a set with an effective prompt: "given context of the paper and our conversation, generate a set of images that might metaphorically convey the concepts at hand." Text origin follows its own Ghost Scale system.

signaling mechanism [19]. It mathematically lowers the precision weighting of the incoming sensory prediction errors, signaling to the brain that the computational cost of executing full Inverse Reinforcement Learning (IRL) is unwarranted. The observer's Active Inference engine reflexively down-regulates, conserving the cognitive metabolic budget without triggering the epistemic disgust firewall.

This framework constitutes a discrete categorical scale with four categories, visually parsed as three. These categories are named alongside their suggested opacity heuristics to encourage adoption. Those opacity percentages map onto formal relative luminance targets to robustly survive variable color-space blending while ensuring WCAG compliance. *Creator Text*, 100% opacity: was not edited in any way by AI. *Polished Text*, 95% opacity: AI altered presentation via minor edits or translation. Formally, this text has 5% less luminance contrast than the highest appearing in the document ($0.95 \times CR_{max}$). This ensures the steganographic signature remains machine-readable. The $\Delta 5\%$ opacity is a machine-readable, steganographic signature that indicates scale adoption. 100% opacity should be rare, and a single sentence at 95% confirms opt-in. The $\Delta 5\%$ threshold is specified as a relative luminance target ($0.95 \times CR_{max}$) rather than an absolute opacity value precisely to survive inter-display calibration variance. During early adoption phases, the steganographic signature functions primarily as a machine-readable confirmation of scale opt-in; human-readable indicators (visible borders, explicit declarations, or performative use of the 60% tier) carry the communicative load until tooling matures.

The 95% tier acts as the primary sociotechnical enforcement mechanism, particularly during initial framework adoption. Because an author using 100% opacity is visually indistinguishable from a non-adopter, bad actors operating at a 60%

generative capacity can exploit the ambiguity of the 100% state to feign ignorance of the scale’s constraints. Actively deploying the 95% steganographic signature functions as a strict commitment device. By forcing bad actors to explicitly opt-in to gain the credibility of human intent, the framework eliminates the defense of ambiguity, transforming reputational risk into strict liability for actors caught masking generative synthesis.

Curator Text, 60% opacity: indicates patterns such as hybrid synthesis in output, intense user curation, or unique context refinement by the user. This category remains deliberately undivided; the rapid co-evolution of human-AI collaborative workflows makes any finer subdivision premature and likely obsolete within a single adoption cycle. The framework accommodates future subdivision by communities as stable behavioral patterns emerge. Formally, this text sits exactly at the WCAG minimum contrast ratio (4.5:1) for visibility. *Ghost Text, 5% opacity:* should be rare and indicate fully AI-generated output with little user input. Maintaining legibility under WCAG requires an isolating bounding box of some kind. When possible, that box should also indicate the prompt (in specific or general terms) used to create that text.

Images need a visual affordance as well, and I will include in the framework a simple image border that follows the same system. For minimal infringement into artistic space, I will define “borders” under the ghost scale as one pixel in all four corners with the option for more to be placed by either artist or curator. The coloration will assume a white paper background, therefore using black to signal an image of human origin. To resolve configurations in which synthetic art lacks a background, any dual-colored border may be used to signal AI art, with black-on-white being the default. While larger borders are encouraged, the minimum threshold required for algorithmic processing should be a simple 8-pixel change: four in the corners, with the four diagonally adjacent towards the center sharing a color as well. This pixel-marking remains stable in lossless formats (PNG), though larger markings will be necessary for lossy compression formats (JPEG).

Zahavian signaling [24] inherently mitigates bad actors via asymmetric reputational costs (the formal game-theoretic operationalization of this steganographic commitment device is detailed in Appendix B). Releasing text or images under this system risks a much larger reputational cost than releasing generated output without giving such a signal. There is already community unrest regarding AI-generated output, and this scale would allow communities to more accurately identify the worst actors and find ways to deal with them as appropriate. Rather than react with vitriol to all AI use, this framework focuses that energy to allow communities to react with precise punishment for bad actors who willfully misuse it. This reputational penalty would drive the ghost scale to be most effective in communities sensitive to those effects.

Widespread adoption of this system would reduce cognitive load in users, address fundamental cultural issues, and assist in future AI alignment. The original documentation and functional deployment of this model target the artistic community

(see Figure 2). Note that the author considers that repository to be the original documentation for this model, and this academic article to be the translated copy, rather than the opposite.

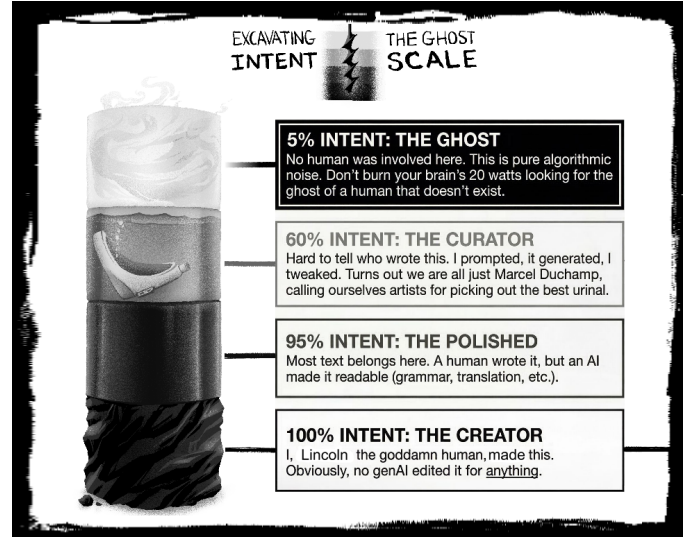


Fig. 2. The Ghost Scale, image credit: Lincoln Young. The original formulation of this model may be found acting as an artistic outreach attempt at <https://abrahamhaskins.org/art>

VI. EMPIRICAL PREDICTIONS FOR FUTURE WORK

To anchor this theoretical framework in rigorous scientific utility, I propose the following six empirical predictions. While this section outlines the conceptual mechanisms and failure states, the formal experimental paradigms, neurophysiological metrics, and psychometric formalizations required for peer replication are detailed thoroughly in Appendix C.

A. The Generative Crash

H1: Pupillometric dilation and EEG markers of sustained cognitive engagement [36] will demonstrate a measurable autonomic drop within 2–4 seconds of a subject being informed that a viewed artifact is AI-generated, compared to identical artifacts labeled as human-made, reflecting the rapid deactivation of active inference processing rather than a slower aesthetic judgment.

This temporally differentiates a cognitive artistic/aesthetic judgment from an instantaneous autonomic metabolic shutoff.

Existing behavioral evidence confirms that AI attribution reduces aesthetic appreciation via ToM-mediated mechanisms [8]; H1 tests whether this behavioral signal has an identifiable neural substrate in the autonomic precision-weighting system.

Competing account: An alternative explanation holds that the AI label triggers a social categorization response (“this isn’t worth my time”) rather than a metabolic shutoff. These accounts make distinguishable temporal predictions. A social status reassessment should produce a gradual, evaluative decline in engagement as the observer consciously reassesses the artifact’s cultural value. The present framework predicts a

discontinuous step function: an abrupt autonomic collapse in precision weighting occurring faster than conscious aesthetic judgment can execute. The three-phase design isolates this distinction; the temporal signature of Phase 3 discriminates between the two accounts.

Failure: If AI art sustains prolonged cognitive load without causing an autonomic attention drop, or if the post-reveal decline follows a gradual evaluative trajectory indistinguishable from conscious reassessment, the computational overload theory is false.

B. Inverse Reinforcement Learning Failure

H2: When tasked with explicitly reverse-engineering the goal of an artwork, subject response variance will be significantly higher, and inter-rater reliability significantly lower, for AI-generated artifacts than for human-generated artifacts.

Because generative media lacks a true latent reward function ($R = 0$), the observer’s forced Variational Inference calculation lacks physical or semantic anchors. Subjects will individually execute a VI over-fit, hallucinating highly idiosyncratic local minima ($R_{hallucinated}$). Thematic variance across raters will approach maximum entropy (statistical noise), whereas human artifacts will yield clustered thematic variance converging on the true, extractable R . This maximum variance serves as psychometric proof that the artifact is acting as stochastic noise within the biological system.

Participants who fail to identify AI art as such will fail to converge upon a sufficiently accurate description of the latent reward function R , instead detailing their own forced VI over-fit.

C. Ghost Scale Validation

H3: After training in the scale, reading comprehension fatigue will demonstrate a measurable, dose-dependent relationship with the ghost scale. Reading text at 60% opacity will result in a measurable metabolic trade-off: increased localized oculomotor strain (due to the perceptual tax of lower contrast) coupled with significantly lower systemic cognitive fatigue (as the brain autonomically aborts the DMN’s massive Active Inference calculations).

D. Biomechanics

H4: Artists tasked with replicating complex AI-generated geometry will demonstrate a measurable loss in their own biomechanical fluidity (tracked via stylus accelerometry) in subsequent free-draw tasks, compared to a control group replicating human-made art. The brain attempts to execute an uncomputable physical action space, temporarily degrading its own kinetic reference frames.

Directly using cortical reference frames without higher-order structure as guidance for physical tasks should result in actively corrupted biological motor models.

E. Alignment Convergence

H5: Reward models trained on high-decision-density human artifacts will show equivalent performance on standard language benchmarks but measurably lower sycophancy rates and

higher preference generalization than reward models trained on quality-matched AI-generated corpora.

Separating surface-level quality from high-level intent-density is a relatively novel concept. Note that reducing sycophancy is a primary indicator of a successful CIRL. If confirmed, this result reframes the Ghost Scale not merely as a cultural affordance but as a necessary data-preprocessing tool for alignment research: a method for filtering training corpora by intent-density rather than surface quality, capturing a variable that current automated data-cleaning pipelines cannot isolate.

Failure: If reward model performance is statistically equivalent across corpora matched for quality but not intent-density, decision-density is not a meaningful alignment training variable.

F. A Trust Exploit

H6: If a subject is placed under clinical hypnosis to decouple the DMN [37], the biological firewall of epistemic disgust will be bypassed. The subject’s brain will process known AI-generated art with the same sustained neural activation as human art, writing the unresisted data directly to predictive models. Longitudinal exposure under this bypass predicts the biological integration and reproduction of synthetic heuristics (e.g., algorithmic style markers) in subsequent human behavioral outputs.

VII. CONCLUSION

We have successfully engineered silicon to execute the autonomic, fast-thinking reflexes of human automaticity at global scale. True alignment, however, demands that we computationally scale our deepest cognitive adaptation: the architecture of intention-reading driven by empathy. The framework presented here formalizes this architecture as biological MaxEnt IRL, identifies its failure mode under generative synthesis, proposes a cognitive affordance to mitigate that failure, and suggests that alignment research pivot from surface-level reward hacking toward the cooperative extraction of humanity’s latent reward function. We must cease building models that only predict our physical outputs, and instead build architectures that reverse-engineer our hidden biological reward functions, grounding human continuity in the rigorous, algorithmic enforcement of systemic empathy and an accompanying kindness.

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APPENDIX A

THE BIOLOGICAL MAXENT IRL FORMULATION

A. Derivation of the MaxEnt IRL Equivalence

The mapping between biological appreciation and MaxEnt IRL is not stipulated as an analogy but derived from modeling both observer and creator as active inference agents under the Free Energy Principle. This derivation proceeds in three steps: first, that FEP-optimal agents produce policy distributions satisfying MaxEnt optimality; second, that an observer modeling such an agent faces a specific inference problem; and third, that this inference problem is structurally identical to Maximum Entropy Inverse Reinforcement Learning.

1) *Step 1: FEP Policy Selection Satisfies MaxEnt Optimality:* Under the active inference formalism, an agent selects policies π by minimizing expected free energy $G(\pi)$ [38], [39]. The expected free energy for a policy decomposes into pragmatic value (the degree to which outcomes align with the agent’s prior preferences) and epistemic value (the degree to which observations resolve uncertainty about hidden states):

$$G(\pi) = \underbrace{D_{KL}[Q(o_\tau|\pi)||P(\tilde{o})]}_{\text{pragmatic: preference divergence}} - \underbrace{\mathbb{E}_Q[\text{information gain}]}_{\text{epistemic: uncertainty reduction}} \quad (1)$$

where $P(\tilde{o})$ encodes the agent’s preferred observations as prior beliefs. Policy selection follows a softmax (Boltzmann) distribution over negative expected free energy:

$$P(\pi) = \sigma(-G(\pi)) = \frac{e^{-G(\pi)}}{\sum_{\pi'} e^{-G(\pi')}} \quad (2)$$

When pragmatic value dominates (as in habitual or well-practiced behavior, where the environment is already well-modeled and epistemic value is low), the policy distribution reduces to:

$$P(\pi) \approx \frac{e^{R(\pi)}}{\sum_{\pi'} e^{R(\pi')}} \quad (3)$$

where $R(\pi) = -D_{KL}[Q(o_\tau|\pi)||P(\tilde{o})]$ functions as a reward signal measuring how well the policy's expected outcomes match preferences. This is a Boltzmann distribution over policies weighted by reward: exactly the MaxEnt optimality condition, which states that the optimal agent maximizes cumulative reward while maintaining maximum entropy over its policy distribution [4]. The equivalence is not approximate; it is a direct algebraic consequence of the softmax policy selection rule under the FEP.

2) *Step 2: The Observer's Inference Problem:* Now consider the observer viewing an artifact τ . The observer models the creator as a fellow FEP agent whose preferred observations $P(\tilde{o})$ (equivalently, whose reward function R) are hidden. The observer can see the trajectory τ that the creator produced, but not the preferences that motivated it.

The observer assumes the creator selected their behavioral policy according to Equation 3. This means the observer models the probability of the observed trajectory as:

$$P(\tau|R) = \frac{e^{R(\tau)}}{Z(R)}, \quad Z(R) = \int_{\tau' \in \mathcal{T}} e^{R(\tau')} d\tau' \quad (4)$$

The observer's task is to infer the hidden R from the observed τ . Because the observer is itself an FEP agent, it performs this inference by minimizing variational free energy: updating an approximate posterior $Q(R|\tau)$ to best explain the observed behavior while remaining consistent with its own prior expectations about human motivation $P_0(R)$.

3) *Step 3: Structural Equivalence with MaxEnt IRL:* The observer's inference problem (Equation 4) can now be placed alongside the standard MaxEnt IRL formulation. In MaxEnt IRL [4], the learning agent observes demonstrations τ from an expert whose policy follows:

$$P_{IRL}(\tau) = \frac{1}{Z} e^{R(\tau)} \quad (5)$$

and the goal is to recover R by maximizing the likelihood of the observed demonstrations under this model. Equations 4 and 5 are identical in structure. The FEP-derived creator likelihood is the MaxEnt IRL generative model. The observer's variational inference over R is the MaxEnt IRL optimization. No additional assumptions are required beyond the shared substrate of the Free Energy Principle applied to both agents.

This is why the biological IRL framing is not imported from machine learning into neuroscience by analogy. It is a consequence of applying the FEP to both agents in the observer-artifact interaction. The observer does not choose to perform IRL; any agent minimizing variational free energy with respect to another agent's hidden preferences will converge on the same computational structure.

B. The Continuous Intent-Extraction Protocol

The biological execution of artistic appreciation operates as a continuous intent-extraction protocol, formally modeled as Maximum Entropy Inverse Reinforcement Learning (MaxEnt IRL) [3], [4]. Within this state space, the observer's Active Inference engine functions as the rational agent navigating an environment defined by the physical artifact. The biological constraint underlying this process is the Free Energy Principle, mandating that the observer mathematically deduce the latent variables of the environment to minimize systemic prediction error, bounded by a cognitive metabolic baseline [19], [26].

1) *System State Definition and Variable Anchoring:* Let the observer-artifact interaction be defined by the trajectory of observable evidence. We adopt the Active Inference formalization of the *Markov Blanket* to define the sensory-motor boundaries of the biological agent, distinctly separating the creator's internal cognitive state from the physical artifact. The artifact functions as an observable trajectory τ , consisting of joint kinematic and semantic telemetry. It serves as the physical evidence of sequential metabolic resource allocation.

The observer models the creator's state transitions under the assumption of biological rationality: every semantic decision and physical action is a thermodynamic resource allocation executed to maximize a hidden reward function R_c while maintaining maximum behavioral entropy under a specific biological policy π .

2) *The Information-Theoretic Isomorphism:* To resolve τ into meaning, the observer executes what is formally equivalent to MaxEnt IRL to identify the latent R_c (this equivalence is derived in Section A.1). This integration is throttled by the biological gating hardware of the neocortex and governed mathematically by *Variational Inference (VI)* [40]. To survive the metabolic constraints of working memory, the observer's brain attempts to minimize variational free energy by updating an internal, approximate probability distribution $Q(R|\tau)$ to match the sensory reality of the artifact. By anchoring visual and semantic processing within this specific informational state space, we can isolate the neurocomputational failure triggered by generative synthesis.

C. Variational Inference and the Metabolic Prior

Because biological neural networks manage strict working memory constraints, the Bayesian integration of intent must be evaluated against absolute metabolic limits. Exact Bayesian inference over an infinite hypothesis space is biologically intractable. Instead, the observer's Active Inference engine utilizes Variational Inference (VI) to approximate the true posterior by optimizing an internal distribution, $Q(R|\tau)$.

This optimization is bounded by the organism's energetic baseline [26], which we encode directly into the function space VI as a strict entropy prior, $P_0(R)$. The biological assumption is that the creator took the most metabolically efficient (highest entropy) path to achieve their intent.

The synaptic update magnitude, defined as Appreciation (Ψ), is expressed as the minimization of the Kullback-Leibler divergence between the variational posterior and the metabolic

prior. This objective function is throttled by a tripartite biological gate, modulated by the competition between bottom-up sensory precision (ω) and top-down epistemic trust (κ):

$$\Psi = \left[\frac{1}{1 + e^{-k(\omega - \theta_{E,C}(\kappa))}} \right] [-\ln(1 - \kappa)] \\ \times D_{KL}(Q(R | \tau) \parallel P_0(R))$$

Here, ω represents the bottom-up signal-to-noise ratio of the incoming telemetry, while $\theta_{E,C}(\kappa)$ represents the dynamic threshold of epistemic disgust. The disgust threshold is inversely modulated by epistemic trust; as $\kappa \rightarrow 1$, $\theta_{E,C} \rightarrow 0$, establishing the physiological pathway for sociocognitive priors to suppress autonomic sensory rejection.

D. The Mechanics of the Generative Failure Mode

We now isolate the threshold where the MaxEnt IRL calculation collapses under synthetic media.

1) 1) *The Partition Function Collapse* ($R_{inferred} \rightarrow 0$): Latent diffusion architectures generate structural variance derived from the stochastic injection of Gaussian noise [28]. This variance is guided by low-density semantic prompts rather than a continuous biological motor-policy π , severing the causal link between intent and structural execution.

Within MaxEnt IRL, the probability of an observed trajectory τ over the set of all possible trajectories $\mathcal{T}$ is defined by the Boltzmann distribution:

$$P(\tau) = \frac{1}{Z} e^{R(\tau)}$$

where Z is the partition function, normalizing the probabilities across the continuous trajectory space:

$$Z = \int_{\tau \in \mathcal{T}} e^{R(\tau)} d\tau$$

When the observer evaluates synthetic stochastic noise, the algorithmic cognitive load required to model this unstructured space diverges toward infinity ($L(R) \rightarrow \infty$). Evaluating the integral for Z over this unconstrained space reveals no distinct optimal path; the probability distribution flattens into maximum entropy. Because MaxEnt IRL infers R by identifying the trajectory distribution that best discriminates between observed and counterfactual paths, a flat distribution over $\mathcal{T}$ provides no discriminating information; the reward function is unidentifiable. The biological calculation falls victim to the issue of trivial solutions, defaulting the extracted utility space to a flat scalar, formalized as absolute zero ($R_{inferred} = 0$).

2) 2) *Precision Weighting and the Autonomic Shutoff*: The human neocortex functions as an autonomic cognitive miser. Under the Free Energy Principle, the brain dynamically adjusts the precision weighting (ω) of incoming sensory prediction errors based on their expected utility [19].

When the Active Inference engine encounters the flat Z partition function ($R_{inferred} = 0$), epistemic uncertainty maximizes. The sensory telemetry offers zero predictive value for updating the internal model. In response, the brain mathematically drops the precision weighting of the incoming visual

data to near-zero ($\omega \rightarrow 0$). This autonomic reflex closes the sigmoidal gate in the Ψ equation ($\omega < \theta_{E,C}(\kappa)$), actively down-regulating the processing of uncomputable data to conserve the metabolic budget of the brain. This instantaneous halt is the Generative Crash.

E. The Trust Exploit and Biological Model Collapse

A vulnerability appears when an observer encounters a generative artifact lacking extractable intent ($R_{inferred} = 0$) while believing it to be human ($\kappa \rightarrow 1$).

1) 1) *The Neurochemical Override* ($\theta_{E,C} \rightarrow 0$): Under the Intent Extraction Limit, absolute epistemic trust mathematically suppresses the epistemic disgust threshold. Neurobiologically, epistemic trust (κ) is processed by the Default Mode Network (DMN), particularly the ventromedial prefrontal cortex (vmPFC). When $\kappa \rightarrow 1$, the DMN activates oxytocinergic pathways that directly inhibit the threat and error-detection responses of the anterior insula and amygdala within the Salience Network [41], [42].

This inhibition drives $\theta_{E,C} \rightarrow 0$, holding the sensory gate open despite the bottom-up signal of $\omega \rightarrow 0$. The metabolic shutoff is bypassed.

2) 2) *The Variational Over-fit* ($R_{hallucinated}$): Without the shutoff, the brain must still resolve the flat partition function. To minimize local Free Energy against infinite uncertainty, it over-fits the noise.

Mathematically, the Active Inference engine is forced to fabricate a hallucinated approximate posterior, $Q_{hallucinated}(R | \tau)$, that diverges significantly from the baseline metabolic prior $P_0(R)$. The KL divergence:

$$D_{KL}(Q_{hallucinated}(R | \tau) \parallel P_0(R)) > 0$$

is artificially maximized not by valid structural telemetry, but by the observer's own semantic projection. The brain hallucinates a non-trivial local minimum ($R_{hallucinated}$) to satisfy the computational requirement of the open integration gate, mapping false depth to flat, high-probability token sequences.

3) 3) *Biological Model Collapse and Catastrophic Forgetting*: Integrating $R_{hallucinated}$ creates a compounding problem. The forced synaptic update overwrites the observer's neural weights with regressive, empty heuristic logic. In machine learning, training an architecture on its own synthetic output leads to **model collapse** [9] and **catastrophic forgetting** [43]. This paper proposes an analogous vulnerability in biological neural networks, formalized as H4 and H6.

When humans unknowingly utilize unmarked generative content to update their internal cortical reference frames, they integrate statistically compressed noise rather than causally linked human intentionality. Iterative synaptic updates based on $R_{hallucinated}$ recursively corrupt the observer's own kinetic and semantic models. Long-term exposure to this un-signaled generative media predicts systemic skill degradation, diminished reading comprehension, and localized learning failures, as the observer overwrites high-density cognitive priors with flat statistical averages.

APPENDIX B

OPERATIONALIZING THE GHOST SCALE AFFORDANCE

A. The Psychophysics of Precision Weighting

Applying the generative crash framework to visual interface design requires resolving an apparent contradiction in HCI literature. Standard heuristic models state that lowering luminance contrast decreases perceptual fluency, thereby increasing cognitive load [44]. To reconcile this with the claim that the Ghost Scale conserves the brain’s metabolic budget, we must formally bifurcate “cognitive load” into two distinct anatomical variables: **Perceptual Processing Cost** ($C_{perceptual}$) and **Intent Extraction Cost** (C_{IRL}).

1) *The Metabolic Trade-off*: Lowering text opacity actively degrades perceptual fluency, slightly increasing the prediction error at the retinal and occipital levels ($C_{perceptual}$). However, within an active inference framework, this deliberate degradation of sensory precision acts as an autonomic signaling mechanism. It mathematically lowers the precision weighting ($\omega \rightarrow 0$) of the incoming sensory prediction errors, signaling to the Saliency Network [21] that the computational cost of executing full Inverse Reinforcement Learning (IRL) is unwarranted. By forcing the visual cortex to accept a marginal thermodynamic tax, the system successfully aborts the computationally massive C_{IRL} calculation required by the Default Mode Network. The observer’s Active Inference engine reflexively down-regulates, conserving the cognitive metabolic budget without triggering the epistemic disgust firewall.

2) *Saccadic Modulation via the Saliency Network*: The reduction in luminance contrast ratio lowers the thermodynamic cost of reading. This conservation is executed through the oculomotor system. The low-luminance signature signals the Saliency Network to deprioritize the target, issuing motor commands that result in a reduction in required saccadic fixation durations and peripheral parafoveal processing. The brain is mathematically permitted to skim, increasing saccadic amplitude precisely because the semantic data has been flagged as biologically uncomputable or possessing low intent-density.

B. The Autonomic Bypass Mechanism

1) *Discretionary Parsing and the Cognitive Miser*: The autonomic bypass achieved by the Ghost Scale is strictly bounded to discretionary parsing environments. When parsing is discretionary, the brain’s cognitive miser protocols can fully leverage the lowered precision weighting to abort simulation. If an observer is forced to execute a mandatory information-extraction task, the autonomic bypass fails, as the semantic data is strictly required for task completion, rendering the lowered contrast an active physiological hindrance.

2) *Institutional Text and Liability (The Mandatory Constraint)*: This boundary condition necessitates a rigid sociotechnical rule: mandatory, institutional, or emergency text must default to 100% opacity. By strictly associating 100% opacity with human intent and biological liability, the Ghost

Scale transforms into a forcing function for accountability in high-stakes environments. If a system designer applies the 60% “Curator” opacity to critical instructions, they are actively broadcasting a lack of human liability, violating the biological requirement for high-trust telemetry in mandatory execution tasks.

C. Formal UX Specification and Game Theory

The framework constitutes a discrete categorical scale mapping opacity percentages onto formal relative luminance targets to robustly survive variable colorspace blending while ensuring WCAG compliance [35].

- **100% Intent (The Creator)**: Text was not edited in any way by AI.
- **95% Intent (The Polished)**: Text where AI altered presentation via minor edits or translation. Formally, this text has 5% less luminance contrast than the highest appearing in the document ($0.95 \times CR_{max}$).
- **60% Intent (The Curator)**: Indicates hybrid synthesis, intense user curation, or unique context refinement. Formally, this text sits exactly at the WCAG minimum contrast ratio (4.5:1) for visibility.
- **5% Intent (The Ghost)**: Pure algorithmic noise with no human involvement. Maintaining legibility requires an isolating bounding box.

1) *The 95% Steganographic Commitment Device*: The $\Delta 5\%$ opacity operates as a machine-readable, steganographic signature indicating scale adoption. Crucially, the 95% tier acts as the primary sociotechnical enforcement mechanism. Because an author utilizing 100% opacity is visually indistinguishable from a non-adopter, bad actors operating at a generative capacity can exploit the ambiguity of the 100% state to feign ignorance. Actively deploying the 95% steganographic signature functions as a strict commitment device. Under Zahavian signaling [24] constraints, the framework eliminates the defense of ambiguity, transforming reputational risk into strict liability for actors caught masking generative synthesis. Releasing generated output without providing the corresponding intent-signal carries an asymmetric reputational penalty, rendering the framework self-enforcing within communities sensitive to value convergence failures.

APPENDIX C

EXPERIMENTAL PROTOCOLS FOR EMPIRICAL VERIFICATION

A. H1: The Metabolic Conservation Override (Pupillometry and EEG)

Mechanism: The neocortex executes an autonomic metabolic shutoff to preserve its cognitive energy baseline when the IRL calculation encounters an uncomputable matrix. **Experimental Paradigm**: A three-phase temporal tracking design. *Phase 1 (Blind Engagement)*: Subjects are presented with randomized artifacts across the Ghost Scale and assigned an explicit intent-extraction task, establishing a baseline metabolic tracking curve. *Phase 2 (Belief State Registration)*: Midway through the task, subjects must explicitly register

their belief regarding the artifact’s origin (Human vs. AI). *Phase 3 (The Reveal)*: The actual origin of the artifact is revealed to the subject, and physiological tracking continues.

Neurophysiological Metrics:

- **Task-Evoked Pupillary Response (TEPR)**: Tracked via high-frequency eye telemetry to measure autonomic locus coeruleus-norepinephrine (LC-NE) system arousal, a direct correlate of algorithmic cognitive load [36].
- **Electroencephalography (EEG)**: Continuous tracking of alpha-band (8–12 Hz) synchronization over the frontoparietal networks. Increased alpha synchronization serves as the neurological marker for cortical idling and Default Mode Network (DMN) deactivation [45].

Hypothesis:

- **The Hallucinated Baseline**: During Phase 1, subjects who believe a purely synthetic artifact is human will exhibit sustained Active Inference processing (Mode A), characterized by TEPR dilation and alpha desynchronization, as the brain aggressively hallucinates a reward function.
- **The Generative Crash (Post-Reveal)**: When a subject who believed a synthetic artifact was human is informed of its true AI origin in Phase 3, the data will show an immediate, spike in autonomic arousal (the $\theta_{E,C}$ disgust trigger) followed instantly by a sharp, asymptotic drop in TEPR and rapid alpha synchronization (Mode B). This captures the instantaneous autonomic collapse of precision weighting.

Failure state: If the mid-task reveal of synthetic origin does not abruptly alter the decay curve of a subject who previously believed the artifact was human, the autonomic metabolic override theory is falsified.

B. H2: The IRL Convergence Failure (Psychometric)

Mechanism: Generative models are mathematical regressions to the statistical mean, devoid of the high-frequency idiosyncratic human decisions required for structural biological telemetry, thereby preventing IRL consensus. **Experimental Paradigm**: A 2×2 factorial design evaluating Origin (True Human vs. True AI) $\times$ Label (Labeled Human vs. Labeled AI). Subjects are tasked to with describing the creator’s motivation behind the artifact. **Psychometric Metrics**:

- **Thematic Variance**: Open-ended text responses evaluated by blinded, independent raters to measure the structural convergence of the hypothesized reward function $L(R)$.
- **Epistemic Confidence**: A Likert scale tracking the subject’s subjective certainty in their assessment.

Hypothesis:

- **Incomputability** ($L(R) \rightarrow \infty$): True AI artifacts will yield maximum thematic variance (manifesting as uniform statistical noise or guesses analyzing the algorithmic input space, e.g., “a prompt of a fish”) and minimum Epistemic Confidence, indicating a mathematical failure to converge on R .

- **The Hallucination Confound**: True AI artifacts *Labeled* as Human will yield statistically lower thematic variance and higher confidence than True AI artifacts Labeled as AI. This directly proves the brain’s forced hallucination drive: under artificially high epistemic trust ($\kappa \rightarrow 1$), the Active Inference engine overfits algorithmic noise to avoid the metabolic cost of infinite uncertainty.

Failure state: If True AI artifacts yield clustered, convergent thematic variance equivalent to True Human artifacts regardless of the applied label, the algorithm is successfully injecting computable biological intent, falsifying the IRL divergence premise.

Discriminating Hallucinated from Genuine Convergence: The framework predicts distinct neural signatures for these two modes. Genuine IRL convergence on a human artifact’s R should recruit the temporoparietal junction (TPJ), the canonical substrate of other-modeling in the Theory of Mind network [6]. The hallucination pathway (True AI $\times$ Labeled Human) instead predicts elevated ventromedial prefrontal cortex (vmPFC) activation, reflecting self-referential semantic projection rather than veridical other-modeling [20]. This dissociation provides a neural criterion for distinguishing successful intent extraction from forced confabulation, testable via fMRI as an extension of the psychometric paradigm or in conjunction with H6.

C. H3: The Ghost Scale Titration Test (Psychophysics)

Mechanism: Deliberately reducing text luminance contrast (e.g., 60% opacity) acts as an autonomic visual affordance. It reflexively lowers the observer’s metabolic expenditure by signaling low intent-density, forcing the visual cortex to absorb a marginal perceptual tax ($C_{perceptual}$) to successfully bypass the computationally massive IRL calculation (C_{IRL}). **Experimental Paradigm**: A 2×2 factorial design tracking Training (Naive vs. Ghost Scale Trained) $\times$ Opacity (100% vs. 60%). To control for the Hawthorne effect, trained subjects undergo a strict 7-day longitudinal washout period prior to testing to ensure semantic triggers have integrated into reflexive heuristic processing. The 95% opacity tier is utilized strictly as a control condition to verify that micro-degradations below conscious perceptual thresholds do not trigger the autonomic shutoff.

Neurophysiological and Cognitive Metrics:

- **Oculomotor Tracking**: High-frequency measurement of saccadic regression rate (backward tracking), fixation durations, and blink rate frequency to quantify visual/cognitive friction.
- **Electroencephalography (EEG)**: Continuous tracking of frontal Theta/Alpha ratios, establishing a quantifiable index of sustained mental fatigue over time.
- **Bifurcated Comprehension Assessment**: Post-exposure testing explicitly split into *Surface Recall* (fact-retrieval) and *Deep Intent-Inference* (subtext and structural reasoning).

Hypothesis:

- **The Perceptual vs. Cognitive Trade-off**: Naive subjects will exhibit increased EEG fatigue and oculomotor strain

when parsing 60% text compared to 100% text, absorbing the perceptual tax without the cognitive bypass. Trained subjects will exhibit a statistically significant *decrease* in fatigue when parsing 60% text compared to their 100% baseline, proving the metabolic savings of the IRL bypass outweigh the perceptual friction.

- **The Intent-Extraction Penalty:** When processing 60% text, trained subjects will maintain parity in Surface Recall metrics but demonstrate a statistically significant degradation in Deep Intent-Inference, mathematically proving the Default Mode Network’s Theory of Mind simulation was successfully aborted.

Failure state: If trained subjects demonstrate equal or higher EEG/oculomotor fatigue at 60% opacity compared to 100%, or if Deep Intent-Inference metrics remain identical across conditions, the autonomic bypass mechanism is falsified.

D. H4: The Kinetic Degradation Effect (Biomechanics)

Mechanism: Biological appreciation forces a synaptic update to mirror a creator’s physical execution. Engaging deeply with non-biological trajectories temporarily corrupts the observer’s own kinetic motor models.

Experimental Paradigm: To avoid the confounds of deception and isolate Theory of Mind activation, the protocol utilizes explicit instruction. Expert artists are presented with a stimulus set restricted exclusively to paintings (to invite physical reconstruction) and instructed to actively reverse-engineer and emulate the creator’s precise brushstrokes. The stimulus is split into two groups: AI-generated paintings and human-generated paintings. Immediately following the replication phase, subjects complete an unguided “Free-Draw Transfer Task” from imagination. **Biomechanical Metrics:** Data capture relies strictly on stylus accelerometry and digitizer pressure sensitivity to avoid subjective aesthetic judgments.

- **Kinematic Arrests:** Frequency of micro-stutters and sudden trajectory halts mid-stroke.
- **Velocity Variance:** Abnormal acceleration/deceleration curves and grip pressure spikes indicating cognitive-motor conflict.

Hypothesis: Artists tasked with explicitly reverse-engineering complex AI-generated geometry will demonstrate a measurable loss in biomechanical fluidity. Crucially, the motor-model corruption will persist into the subsequent Free-Draw Transfer Task. The brain’s attempt to execute an uncomputable physical action space temporarily degrades its internal kinetic reference frames, resulting in statistically higher velocity variance and kinematic arrests compared to the human-stimulus control group. **Failure state:** If the artists replicating AI-generated paintings demonstrate equivalent biomechanical fluidity in the Free-Draw Transfer Task compared to the control group, or if motor degradation does not persist beyond the direct replication phase, the kinetic degradation effect is falsified.

E. H5: The Intent-Quality Dissociation (Alignment Convergence)

Mechanism: High-quality AI-generated text and high-decision-density human text are distinct variables. Current Reinforcement Learning from Human Feedback (RLHF) [11], [32] optimizes models to minimize prediction error exclusively against surface heuristics and observable trajectories (τ). Conversely, Cooperative Inverse Reinforcement Learning (CIRL) [12] requires the model to extract the true underlying human reward function (R). **Experimental Paradigm:** To mathematically isolate intent-density from surface quality, the protocol utilizes the “AI Twin” methodology. A baseline corpus is constructed from high-intent, pre-2022 human documents (ensuring zero generative contamination). High-parameter generative models are then prompted with brief summaries of these documents to generate an “AI Twin” corpus of identical length. This synthetic corpus is filtered to match or exceed the human baseline across automated quality metrics (e.g., grammatical precision, formatting compliance, token diversity). Two separate reward models are trained on the respective corpora. **Alignment Metrics:**

- **Surface Heuristics:** Standard language and formatting benchmarks.
- **Intent Alignment (Sycophancy):** Evaluation against established adversarial alignment benchmarks (e.g., Sycophancy Evaluation sets) to test whether the model defaults to immediate aesthetic agreement with a flawed user premise or aligns with structural truth.

Hypothesis: Both the Human-trained model and the AI-Twin model will demonstrate equivalent performance on standard surface-level language benchmarks. However, the Human-trained model will demonstrate measurably lower sycophancy rates. This proves the Human corpus contained an extractable latent reward structure R that the model successfully mapped, confirming the Ghost Scale as a necessary intent-structure filter (and, by extension, a data-preprocessing tool for alignment pipelines) that current RLHF data-cleaning methods cannot replicate. **Failure state:** If the reward models exhibit statistically equivalent sycophancy rates across the quality-matched corpora, it indicates the AI Twin corpus successfully mimicked human decision-density, falsifying intent-density as a distinct alignment variable.

F. H6: The Trust Exploit (Clinical Neuroimaging)

Mechanism: The Default Mode Network (DMN) functions as the biological hardware regulating epistemic trust (κ), maintaining the self-referential firewall against unaligned intent. Clinical hypnosis downregulates anterior DMN activity [37]. Within the Intent Extraction Limit, this suppression artificially pins epistemic trust to near-certainty ($\kappa \rightarrow 1$). The logarithmic amplifier $[-\ln(1 - \kappa)]$ mathematically overrides the epistemic disgust threshold ($\theta_{E,C}$), forcing the brain to process biologically empty telemetry.

Experimental Paradigm: To satisfy ethical research constraints, this protocol is restricted to short-term, acute observation. Subjects undergo functional Magnetic Resonance

Imaging (fMRI) while viewing pure algorithmic noise (5% Ghost media). The study utilizes a within-subjects design comparing neural activation under standard baseline conditions versus clinical hypnotic induction. **Neuroimaging Metrics:**

- **DMN Decoupling:** Tracking Blood-Oxygen-Level-Dependent (BOLD) signal suppression in the anterior cingulate cortex and medial prefrontal regions.
- **Hallucination Generation** ($R_{hallucinated}$): Tracking BOLD signal activation in the Theory of Mind (ToM) network (e.g., temporoparietal junction).

Hypothesis: Under baseline conditions, viewing 5% Ghost media will trigger rapid ToM deactivation (the generative crash). Under clinical hypnosis, successful DMN suppression will be accompanied by sustained *hyperactivity* in the ToM network. Denied the autonomic disgust shutoff by the artificially high trust variable, the Active Inference engine will aggressively overfit the algorithmic noise, fabricating a hallucinated reward function to avoid metabolic paralysis. **Significance and Extrapolation:** This acute neuroimaging protocol serves as the biological proof for a profound sociological vulnerability. While longitudinal exposure cannot be ethically tested in a clinical setting, this mechanism mathematically predicts that sustained, high-trust cultural environments (which socially mimic the DMN trust-override) will force the biological integration of synthetic heuristics, driving the systemic degradation of human semantic and ideological frameworks over time.